



Tengizchevroil

Heavy Duty Vehicle and Equipment Management Procedure

Тенгизшевройл

Процедура по управлению большегрузными транспортными средствами и оборудованием

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Process Overview

Scope and Purpose

This procedure applies to the management and utilization of TCO-owned and rented heavy-duty vehicles (HDV) and heavy-duty equipment (HDE). It encompasses all processes related to equipment requests, booking, mobilization, utilization monitoring, demobilization, and payment.

The procedure ensures the efficient allocation of resources, cost optimization, and alignment with TCO's strategic objectives for operational excellence.

The scope includes:

- Integrating a centralized database for equipment management
- Streamlining equipment allocation and sourcing through digital decision-making tools
- Establishing clear roles and responsibilities for stakeholders involved in the process
- Monitoring utilization to improve efficiency and reduce operational costs
- Ensuring compliance with safety, maintenance, and inspection standards
- The policy applies to all TCO departments and Business Partners (BPs) involved in the usage, booking, monitoring, and maintenance of HDVs and HDEs within TCO operations

Purpose

The purpose of this procedure is to provide a structured and transparent framework for managing HDV/HDE across TCO operations. It aims to:

- Enhance the efficiency and reliability of equipment utilization
- Ensure safe and timely mobilization of resources to meet operational demands
- Automate and simplify decision-making processes to minimize delays and improve resource allocation
- Reduce operational costs by optimizing the use of TCO-owned equipment and minimizing reliance on rented resources
- Integrating performance metrics to increase utilization efficiency

Definitions

Heavy Duty Vehicle (HDV) - a vehicle or truck designed for transportation of cargoes and with the maximum technically allowable weight greater than 3.5 tons

Heavy Duty Equipment (HDE) – self-propelled heavy machinery / equipment specifically designed for construction activities, lifting and earthmoving tasks (Wheeled / stationary)

Heavy Duty Vehicle (HDV)	Heavy Duty Equipment (HDE)
Trucks (boom, chemical, dump, vacuum, fire, flatbed, fuel, steam, water, refrigerator etc.)	Bobcat, excavator, forklift, loader, tractor, manlift, trailer, compressor, generator, heater, pump other equipment

Linkages to Other Business Processes and Resources

1. TCO MVS procedure SP-35
2. FMS guidelines
3. TCO Policy CON.07.01
4. Asset Disposal procedure
5. Finance inventory process
6. Vehicle Technical Inspection ([Business Partners Fleet Inspection](#))
7. Vehicle Access Management System (PAMS)

Resources, Roles and Responsibilities

The following table outlines the roles and responsibilities associated with this process.

Role	Responsibilities
Fleet Owner (TCO) (BP)	Manage owned HDV/HDE, ensure availability, maintenance, and operational readiness • Maintain an up-to-date inventory of all HDV/HDE, including operational status and location, and details/specifications, photos of an asset. • Define criteria for each HDV/HDE - assigned, shared, shared with conditions. • Approve or reject requests from TCO representatives based on operational priorities and existing commitments. • Schedule and coordinate mobilization and demobilization of equipment as per approved requests. • Perform regular maintenance and inspections according to manufacturer recommendations.

Role	Responsibilities
	- Update status for each step of the equipment's lifecycle in the digital management system. - Ensure equipment is mobilized/replaced on time to meet Requestor project schedules. - Ensure contract information are correctly indicated for BP's fleet (BP)
Requestor	Submit accurate and complete booking requests by equipment type or specification, dates required, location of use, cost center - Daily monitor fleet utilization - Timely revise request as per business need
Process Owner - (TCO Logistics)	- TCO owned fleet maintenance, registration, asset disposal process, BP's fleet inspection to provide access to TCO facilities - Monitor the timely revision of procedure, conduct change requests, assign roles, administrate the platform performance - Provide utilization analysis to stakeholders - Maintain IoT telematics performance
Contract team	- Ensure fair and competitive agreements with BPs. - Timely creates/revise SR/SOs in SMART GEP, and confirms in the HDV/HDE system - Maintain up-to date catalog costs in the system
Approvers	Validate requests based on business needs and budget alignment.
Cost Controller	

Procedure

1. Centralized Database

- 1.1. All HDV/HDE and attachments (including BP's) must be registered in the centralized database with unique IDs, specifications, photos and ownership status.
- 1.2. Real-time database updates on equipment availability, location, and utilization are mandatory.
- 1.3. IoT/Telematics services must be installed on all TCO and BP HDV/HDE as per owner's requests for monitoring.
- 1.4. IoT/Telematics services expenses must be covered by fleet owner

2. Booking and Allocation

- 2.1. All equipment requests must be processed through the digital booking platform. No request in the system - no vehicle/equipment.
- 2.2. The system will prioritize TCO-owned assets. If unavailable, BP rentals will be suggested based on a pre-set decision matrix (cost, reliability, availability).
- 2.3. Equipment classification (e.g., type, capacity, location) must guide selection.
- 2.4. TCO owned HDV/HDE categories:
 - Shared – HDV/HDE are provided to requestor without any condition as per requested dates.
 - Shared with conditions – HDV/HDE are provided to requestor, but owner can reallocate the resources for high priority requests. Requestor must not 100% rely on this category of vehicles/equipment.
 - Assigned – HDV/HDE is assigned to one facility or location and cannot be shared to other users due to emergency support or daily usage.
- 2.5. Rented HDV/HDE categories:
 - Shared with conditions – BP has an open long-term SR for HDV/HDE and can be shared to other requestors as per requested date. (For example, drilling HDV/HDE - other departments can request AGI HDV/HDE if drilling has no open works (no payment is required from other departments, MSS crane SRs – requester department will pay as per requested dates, no new SR is required)
 - Assigned – HDV/HDE is assigned to one facility or location and cannot be shared to other users due to emergency support, daily usage or specific.
- 2.6. Priority requests:

- P1 – Critical works impact to Production, emergency and TA
- P2 – Planned and routine production side works
- P3 – non-production side works
- P4 – other works

3. Utilization Monitoring

- 3.1. Utilization rates must be tracked daily by user department using IoT/Telematics services, motor hours, mileage and other applicable parameters based on working specific, such as crane boom movement, vacuum truck pump, etc. The following formula will measure monthly utilization: $\text{Actual motor hours and mileage days} / \text{total hours in month}$ (as a default 12 hours per day but can be adjusted to 24 hours as per request).
- 3.2. Review of company HDV/HDE utilization must be conducted by owners on regular basis.
- 3.3. Underutilized equipment has to be flagged for reassignment, demobilization or decommissioning.
- 3.4. Utilization monitoring excludes emergency HDV/HDE which are pre-determined and approved

4. Requirements for IoT and Telematics

4.1. TCO

5. Maintenance and Inspection

- 5.1. All HDVs and HDEs must have a maintenance program in accordance with the recommendations of the Manufacturer and RoK regulations.
- 5.2. TCO owned HDV/HDE maintenance is performed by specialized Business Partners.
- 5.3. Operators must complete initial and periodic training for safe equipment operation as per existing TCO policies.
- 5.4. Daily pre-operation safety checks are mandatory for all assets.

6. Inventory and write-off

- 6.1. Owners must conduct annually inventory physical checks of all their HDV/HDEs under their respective Cost Center (CC) and AFE
- 6.2. CC or AFE changes must be approved by owners from both sides and as per act of acceptance and notification to Logistics group to update the system
- 6.3. Owner is responsible for the write-off process based on technical evaluation and follow: TCO Policy CON.07.01 Disposal of Surplus Asset and Material
- 6.4. All transactions must be stored in the system for historical data tracking

7. Cost Management

- 7.1. The decision matrix must factor in total costs, including rental rates, operator fees, and fuel expenses.
- 7.2. Rented HDV/HDEs must be paid by unique asset ID in the system to avoid overbilling and cross checking with IoT data (geolocation, actual worked hours)
- 7.3. Actual contracts spend for HDV/HDEs must be tracked by categorized description of vehicles and equipment
- 7.4. HDV/HDE utilization scorecard must be consolidated and shared with department leadership on a quarterly basis.

8. Compliance

- 8.1. Department owners of HDV/HDE must ensure respective standard operating procedures support this guideline
- 8.2. Regular audits will ensure compliance with operational, financial, and safety requirements

See **Attachment 1** for the visual reference to the Process Map below: TBD

Process Flow Diagram - [HDVHDE_process_flow_20.03.vsd](#)

Measurement

Leading metrics:

- TCO HDV/HDE Utilization rate, benchmark Utilization>70%
- No gaps in Inventory

Lagging metrics:

1. Actual Spend \$
2. Total Fleet decrease
3. CAPEX decrease
4. # of missing HDV/HDE -> 0

Continuous Improvement

Document Control Information

Description	Procedure
Revision Date	February 2025
Revision Frequency	TBD
Control Number	

Document List

Process Flow Diagram

Reference materials

Appendix 1

Detailed Instruction Procedure

The procedure consists of the following steps starting from the time when Requester need to search and book the HDV/HDE:

Step	Role	Description	System	Instructions
1	Requestor	Define Business Need	Out of system	- Requester knows requirements and needs to start searching for HDV/HDE - Requester opens the HDV/HDE Management tool
2	Requestor	JDE WO with specific Work Center for logistics shared services (CRANE, DTRK, FTRK, FORKL)	JDE	- JDE integrated with HDV/HDE system - Requester can optionally select WO number from drop down list
3	Requestor	Request creation Requester can add several HDV/HDE to one request	HDV/HDE system	- Manually created by requestor with defined business need - Created by JDE with specific work center - Assign unique number for request id
3	System	Generate system request ID	HDV/HDE system	
4	Requestor	Search page	HDV/HDE system	Requester inputs required search criteria for HDV/HDE in the Search page of the HDV/HDE Management tool
5	System	System display page (*each HDV/HDE owner is responsible for managing data in proper and actual condition)	HDV/HDE system	System provides a list of available HDV/HDE with tag criteria (shared, shared with condition, assigned)
6	Requester	Selecting TCO Owned HDV/HDE as priority	HDV/HDE system	According to HDV/HDE Policy, Requester must

				first consider using TCO Owned assets, and if not available, then consider BP Owned assets to rent them
7	Requester	Finalize request	HDV/HDE system	- Add additional HDV/HDE - If required night shift driver/operator
8	Requester	Submit Request	HDV/HDE system	Press Submit button
9	TCO HDV/HDE Owner (Cross Functional)	Confirm	HDV/HDE system	- Confirm readiness of HDV/HDE to mobilize - If not available, the System opens to Requester a Search page
10	Requester	Receives confirmation	HDV/HDE system	Requester receives confirmation notification if HDV/HDE Owner confirms the availability of HDV/HDE requested
11	TCO HDV/HDE Owner (Cross Functional)	Prepare for mobilization	Out of system	- Prepare work task for drivers/operator - Obtain required accesses to Facility
12	Driver/Operator	Mobilize Vehicle/Equipment	Out of system	Prepare work task for drivers/operator
13	Driver/Operator	Execute safe work	Out of system	
13	Requester	Review Utilization	HDV/HDE System Power BI	Requester reviews daily utilization of ordered HDV/HDE by IoT telematics
14	Requester	(Optional) Extension of request	HDV/HDE system	Based on utilization and confirmation from end users, requester can extend the order for HDV/HDE based on availability

16	Requester	Change request if demob is before end date	HDV/HDE system	Based on utilization and confirmation from end user, requestor can initiate release date earlier than requested date
17	Driver/Operator	Complete/Demobilize Vehicle/Equipment.	Out of system	- Completion of work - Demobilize the HDV/HDE
18	System	Close request in the system with end date	HDV/HDE system	
19	System	HDV/HDE available in the system	HDV/HDE system	
20	Requestor	Feedback	HDV/HDE system	Provide feedback for the service executed
If Non TCO-owned HDV/HDE is selected				
21	System	Shows BP list by category, sorted by recommended and with existing SRs (cranes, drilling vehicles etc.)	HDV/HDE system	Scoring system based on weighted coefficient: - Cost – hourly/daily, monthly rates - Reliability – using historical data, availability, cancellation, downtime, replacement etc. - Feedback score – received after work completion
22	Requester	Provide justification submit request for rented HDV/HDE	HDV/HDE system	Add comment in the system that confirms TCO-owned HDV/HDE is not available, and the work cannot be postponed
23	CC Owner or Delegated Person	Approve or Reject	HDV/HDE system	If rejected, then system shows updated list of HDV/HDE to select
24	BP HDV/HDE Owner	Confirms availability	HDV/HDE system	If not confirmed, then system shows updated list of HDV/HDE to select

25	Requester	Receive confirmation from CC owner, BP	HDV/HDE system	•
26	Requester	Create RWA in DCT if required	DCT	•
27	SCM Contract Specialist	Receive request by email from the system (SCM Central Team)	HDV/HDE system	•
28	SCM Contract Specialist	Create SR/SO in SMART GEP if required	SMART GEP	•
29	SCM Contract Specialist	Confirms SO Creation in SMART GEP	HDV/HDE system	•
30	BP HDV/HDE Owner	Prepare for mobilization	Out of system	• Prepare work task for drivers/operator • Obtain required accesses to Facility • Ensure SR/SO is open
31	Driver/Operator	Mobilize Vehicle/Equipment	Out of system	• Prepare work task for drivers/operator
32	Driver/Operator	Execute work	Out of system	•
33	Driver/Operator	Complete/Demobilize Vehicle/Equipment	Out of system	• Completion of work • Demobilize the HDV/HDE
34	Requester	Review Utilization	HDV/HDE system	• Requester reviews daily utilization of ordered HDV/HDE by IoT telematics
35	Requester	Change status if demob is before end date	HDV/HDE system	• Based on utilization and confirmation from end user, requestor can initiate release date earlier than requested date
36	HDV/HDE Owner BP	Order change	HDV/HDE system	• Confirms order change • Confirms availability and/or initiate demobilization
37	HDV/HDE Owner BP	Close request	HDV/HDE system	• Confirms dates for the service provided

				- Confirm closing the request - Initiate demobilization
38	System	Close request in the system with end date	HDV/HDE system	•
39	System	Close request in the system	HDV/HDE system	•
38	System	HDV/HDE available in the system	HDV/HDE system	•
39	Requester	Submit feedback	HDV/HDE system	Provide feedback for the service executed
41	System	Calculates timesheet in DCT	HDV/HDE system	•
42	System	Calculates no overbilling with HDV/HDE ID and auto populate with telematics data for each day	HDV/HDE system	•
43	System	Confirm AoA	AoA system	Automatically cross check actual worked hours by IoT telematics data
44	Cost controller	Confirm timesheet in SMART GEP	SMART GEP	•
45	System	Payment processed	SMART GEP	•
Process for Managing HDV/HDE database				
1	TCO HDV/HDE Owner (Cross Functional)	Create New / Update HDV/HDE records		•
2	BP HDV/HDE Owner	Create New / Update HDV/HDE records		•
3	System	Update database		•
4	System / Owner	Database is updated message		•